

E7 Introduction to Computer Programming for Scientists and Engineers

Lecture week 11-B Integration

A few applications of integration

- **Mechanics**

- Find centroids and moments of inertia of a body
- Find total forces acting on a body

- **Kinematics**

- Obtain velocities from accelerations and displacements from velocities

- **Signal processing**

- Calculate the energy of a signal

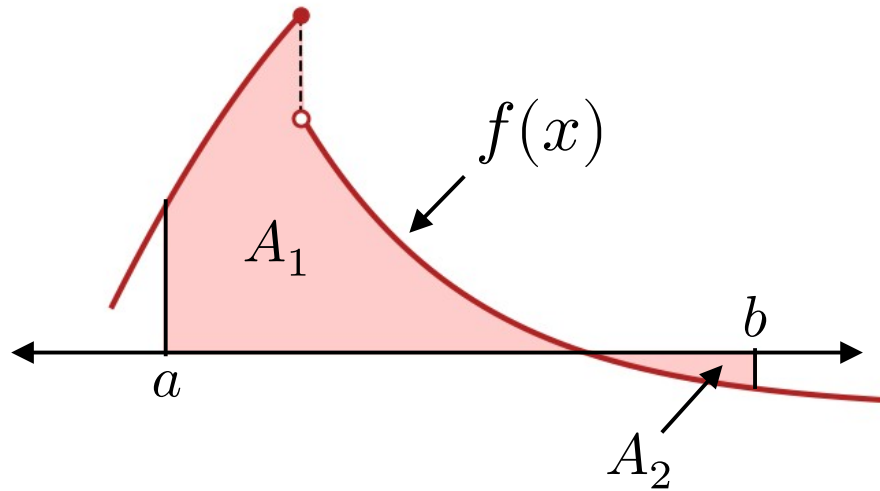
- **Hydrology**

- Find the total volume of a body of water

- ...

Definition of the integral

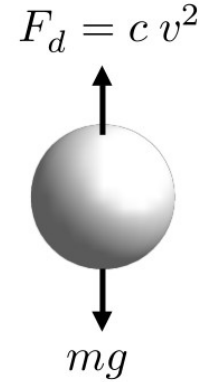
$\int_a^b f(x) dx$... signed area between $f(x)$ and the x axis.



Example: Find the instant at which a spherical object, dropped from a height of 15 meters, will hit the ground.

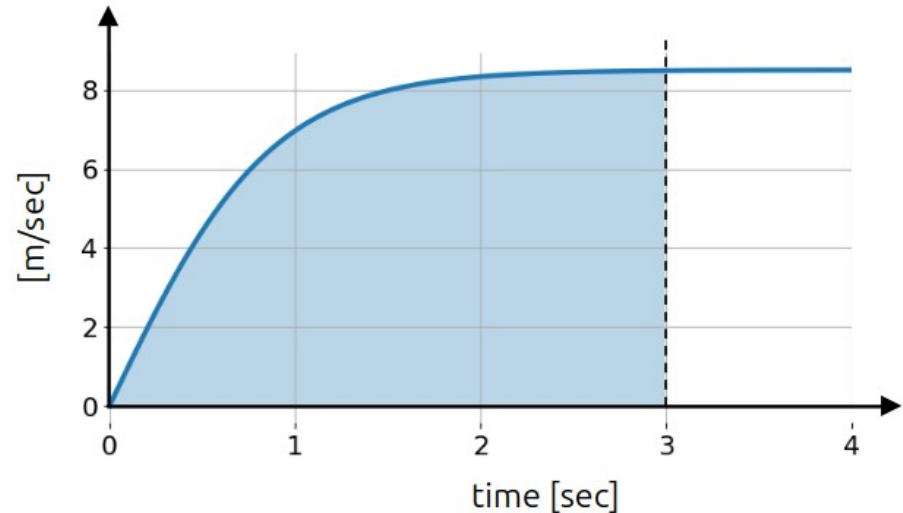
1) Apply Newton's second law:

$$mv'(t) = cv^2(t) - mg$$
$$v(0) = 0$$



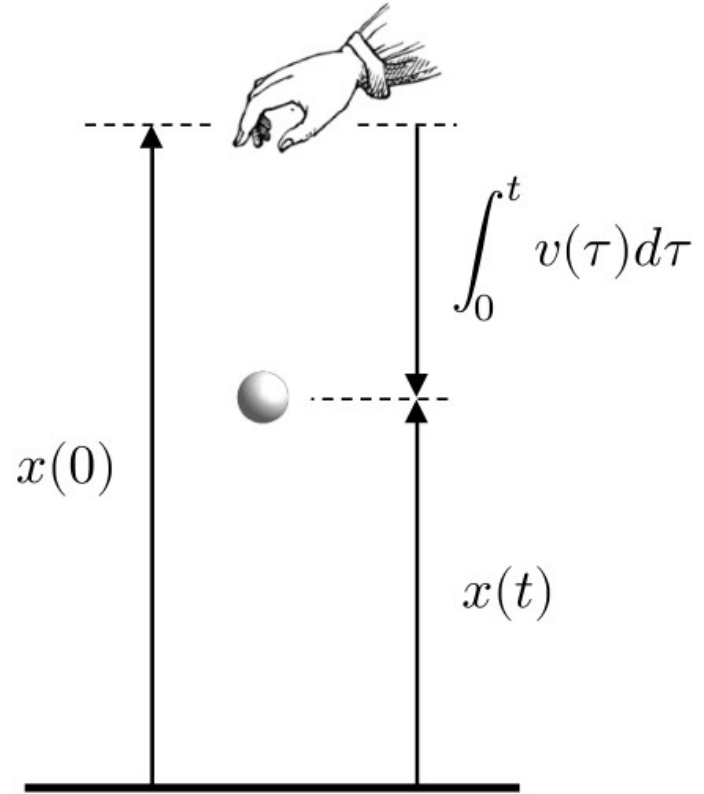
2) Solve ODE for the velocity:

$$v(t) = -\sqrt{\frac{gm}{c}} \tanh \left(\sqrt{\frac{gc}{m}} t \right)$$



3) Integrate the velocity to obtain position:

$$x(t) = x(0) + \int_0^t v(\tau) d\tau$$

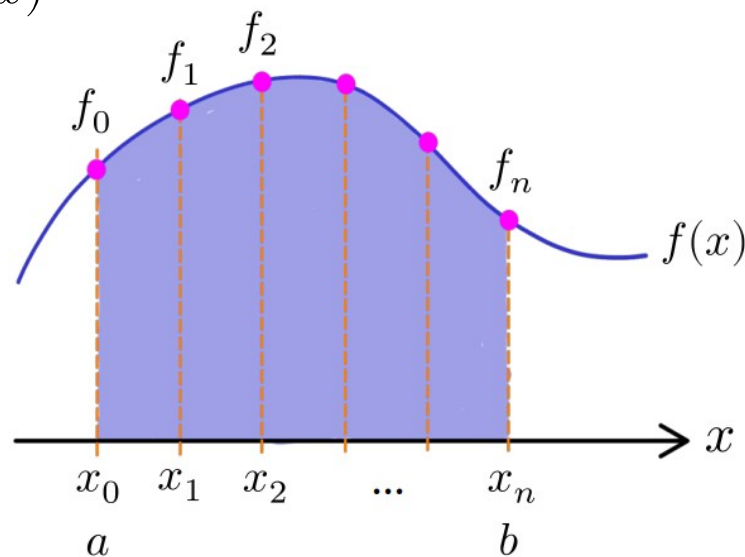


Newton-Cotes formulas

To evaluate $I = \int_a^b f(x) dx$

1. Split $[a, b]$ into n segments of width h
2. Approximate $f(x)$ using a polynomial $p(x)$
3. Integrate $p(x)$.

$$\tilde{I} = \int_a^b p(x) dx \approx I$$



Finding $p(x)$

- $p(x)$ is the simplest polynomial that goes through all $n + 1$ points.
- It is called the *Lagrange polynomial* and it is easy to compute.

$$\ell_i(x) = \frac{(x - x_0)}{(x_i - x_0)} \cdots \frac{(x - x_{i-1})}{(x_i - x_{i-1})} \frac{(x - x_{i+1})}{(x_i - x_{i+1})} \cdots \frac{(x - x_n)}{(x_i - x_n)} \quad \forall i \in \{0, \dots, n\}$$

$$p(x) = \sum_{i=0}^n f_i \ell_i(x)$$

Notice that:

$$1. \ell_i(x_i) = 1 \quad \forall i \in \{0, \dots, n\}$$

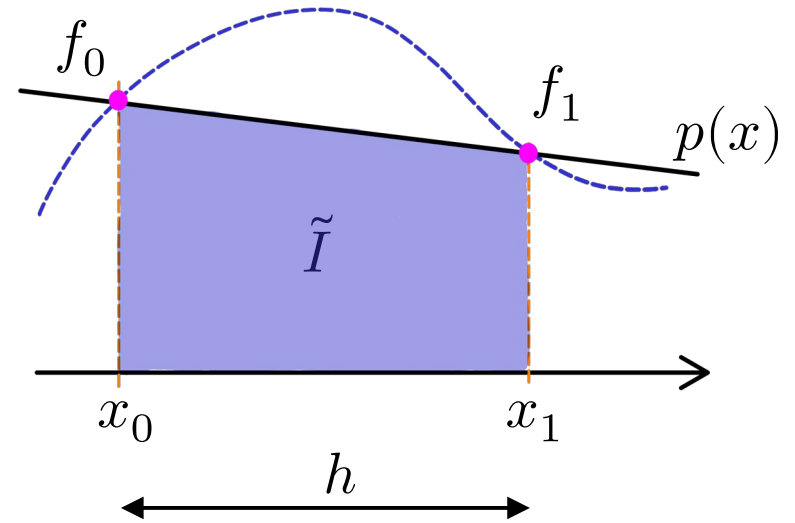
$$2. \ell_i(x_j) = 0 \quad \text{if } i \neq j$$

$$3. \text{order}(\ell(x)) = n \Rightarrow \text{order}(p(x)) \leq n$$

First order Newton-Cotes: The trapezoidal rule

$$p(x) = \frac{f_1 - f_0}{h} x + \frac{f_0 x_1 - f_1 x_0}{h}$$

$$\tilde{I} = \int_{x_0}^{x_1} p(x) dx = \frac{f_0 + f_1}{2} h$$



Error of the trapezoidal rule

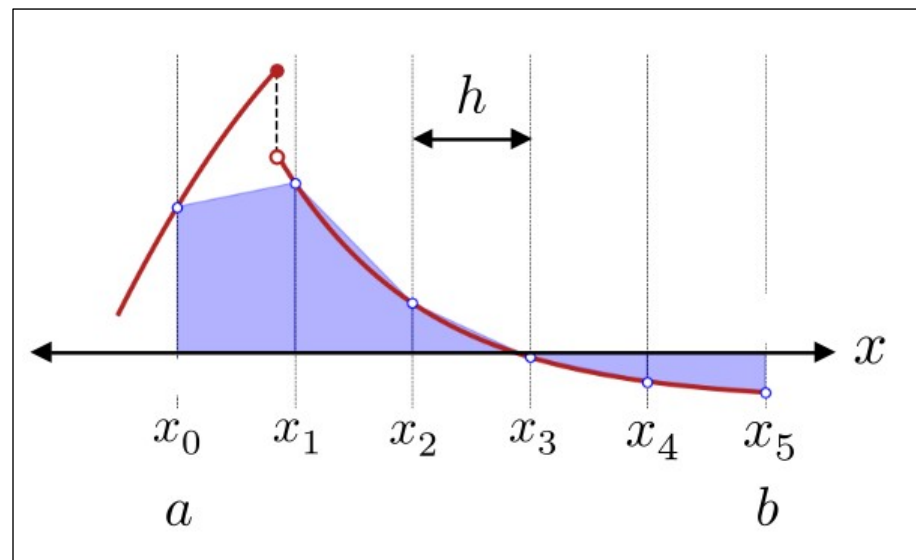
- If
1. f is continuous on the interval $[a, b]$
 2. f has bounded second derivative: $M_2 = \max_{x \in [a, b]} |f''(x)|$

Then
$$|I - \tilde{I}| \leq \frac{(b-a)^3}{12} M_2$$

Composite trapezoidal rule

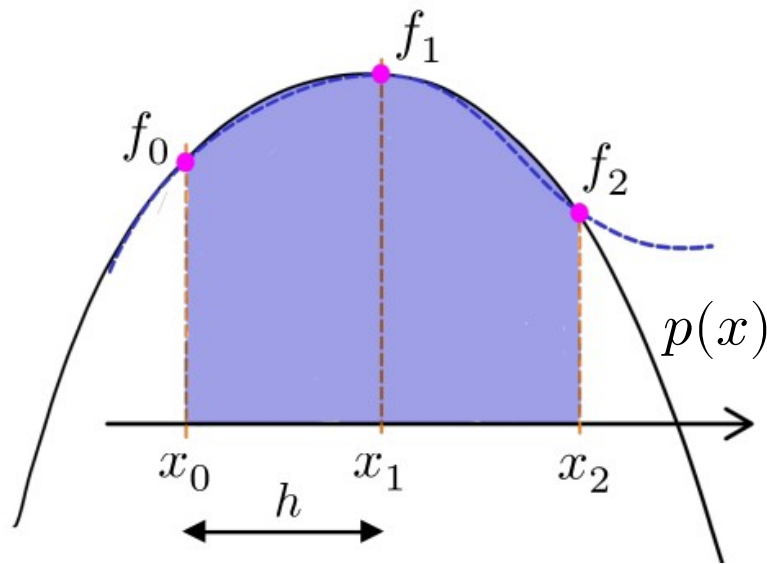
- Apply the trapezoidal rule to each segment.
- Accumulated error:

$$\begin{aligned} |\text{total error}| &= \sum_{i=1}^n |\text{error } i'\text{th segment}| \\ &\leq n \frac{h^3}{12} M_2 \\ &= \frac{(b-a)h^2}{12} M_2 \end{aligned}$$



Second order Newton-Cotes: Simpson's rule

$$p(x) = f_0 \frac{(x - x_1)}{(x_0 - x_1)} \frac{(x - x_2)}{(x_0 - x_2)} + f_1 \frac{(x - x_0)}{(x_1 - x_0)} \frac{(x - x_2)}{(x_1 - x_2)} + f_2 \frac{(x - x_0)}{(x_2 - x_0)} \frac{(x - x_1)}{(x_2 - x_1)}$$



without loss of generality, take $x_0 = -h$, $x_1 = 0$, $x_2 = h$

$$\begin{aligned} p(x) &= f_0 \frac{x}{h} \frac{(x-h)}{2h} + f_1 \frac{(x+h)}{h} \frac{(x-h)}{-h} + f_2 \frac{(x+h)}{2h} \frac{x}{h} \\ &= \vdots \\ &= \frac{1}{2h^2} \left((f_0 - 2f_1 + f_2) x^2 + (f_2 - f_0) h x + 2f_1 h^2 \right) \end{aligned}$$

$$\tilde{I} = \int_{-h}^h p(x) dx = \dots = \frac{h}{3} (f_0 + 4f_1 + f_2)$$

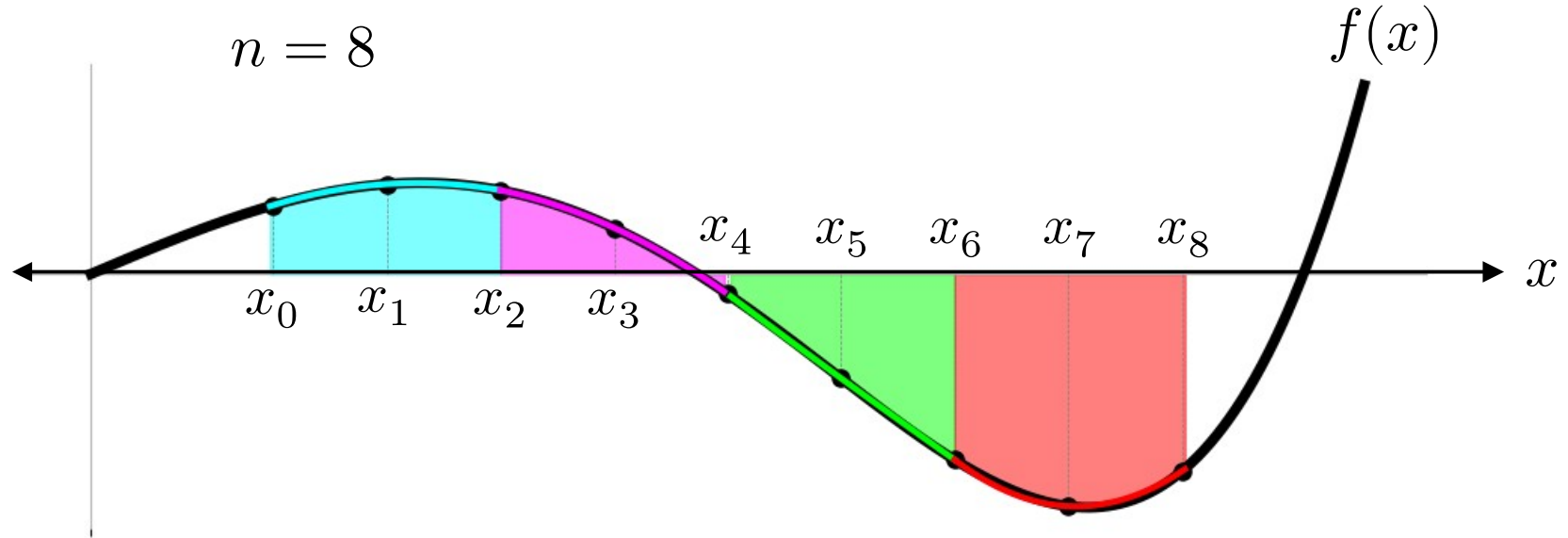
Error of Simpson's 1/3 rule

If 1. f is continuous on the interval $[a, b]$

2. f has bounded fourth derivative: $M_4 = \max_{x \in [a, b]} |f^{(4)}(x)|$

$$\text{Then } |I - \tilde{I}| \leq \frac{(b-a)^5}{2880} M_4$$

Composite 1/3 rule



$$|\text{total error}| \leq \frac{(b-a)h^4}{2800} M_4 \quad \dots O(h^4)$$